

Physics Problem Solutions

Mechanics: Center of Mass & Dynamics

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1 Problem 9: Center of Mass of 3 Particles

1. System Setup & Givens

Reference Frame: We choose the origin O ($x = 0$) at the position of the first particle (m_1). The positive x -axis extends to the right.

- Since all particles lie on this axis, the ordinate of each particle is zero ($y_i = 0$). Consequently, the ordinate of the Center of Mass is also zero ($y_G = 0$).

List of Givens:

- **Particle 1:** $m_1 = 10$ g at $x_1 = 0$ cm
- **Particle 2:** $m_2 = 20$ g at $x_2 = 50$ cm
- **Particle 3:** $m_3 = 30$ g at $x_3 = 50 + 50 = 100$ cm

2. Calculation

The abscissa of the center of mass G is given by:

$$x_G = \frac{\sum m_i x_i}{\sum m_i} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3}$$

Substituting the values:

$$x_G = \frac{(10 \cdot 0) + (20 \cdot 50) + (30 \cdot 100)}{10 + 20 + 30}$$

$$x_G = \frac{0 + 1000 + 3000}{60} = \frac{4000}{60} = \frac{200}{3}$$

3. Final Result

Converting to decimal:

$$x_G \approx 66.67 \text{ cm}$$

Interpretation: The Center of Mass is located between the 20 g mass (at 50 cm) and the 30 g mass (at 100 cm), specifically 16.67 cm to the right of the middle mass.

2 Problem 14: Center of Mass of Rod Systems

General Method: Since the rods are uniform and homogeneous, the Center of Mass (CoM) of each individual rod is located at its geometric midpoint. We treat each rod as a particle of mass m located at this midpoint, then apply the system CoM formula.

Case a: L-Shape System

Two rods of mass m and length L joined at 90° . Let the junction be the origin $(0, 0)$.

- **Rod 1 (Horizontal):** Along x -axis. Midpoint $G_1(\frac{L}{2}, 0)$.
- **Rod 2 (Vertical):** Along y -axis. Midpoint $G_2(0, \frac{L}{2})$.

Calculation ($M_{total} = 2m$):

$$x_G = \frac{m(L/2) + m(0)}{2m} = \frac{L}{4}, \quad y_G = \frac{m(0) + m(L/2)}{2m} = \frac{L}{4}$$

Result: $G\left(\frac{L}{4}, \frac{L}{4}\right)$ (Lies on the bisector).

Case b: V-Shape System (Angle α)

Rod 1 on x -axis, Rod 2 makes angle α with Rod 1 at the origin.

- **Rod 1:** Midpoint $G_1(\frac{L}{2}, 0)$.
- **Rod 2:** Midpoint $G_2(\frac{L}{2} \cos \alpha, \frac{L}{2} \sin \alpha)$.

Calculation:

$$x_G = \frac{m(L/2) + m(\frac{L}{2} \cos \alpha)}{2m} = \frac{L}{4}(1 + \cos \alpha)$$

$$y_G = \frac{m(0) + m(\frac{L}{2} \sin \alpha)}{2m} = \frac{L}{4} \sin \alpha$$

Result: $G\left(\frac{L}{4}(1 + \cos \alpha), \frac{L}{4} \sin \alpha\right)$

Verification: If $\alpha = 90^\circ$, $\cos 90 = 0$, $\sin 90 = 1$. The result becomes $(\frac{L}{4}, \frac{L}{4})$, matching Case a.

Case c: U-Shape System (3 Rods)

Two vertical rods connected by a horizontal rod. Origin at bottom-left corner.

- **Left Vertical:** $G_1(0, \frac{L}{2})$.
- **Bottom Horizontal:** $G_2(\frac{L}{2}, 0)$.
- **Right Vertical:** $G_3(L, \frac{L}{2})$.

Total Mass $M = 3m$.

$$x_G = \frac{m(0) + m(L/2) + m(L)}{3m} = \frac{1.5mL}{3m} = \frac{L}{2}$$

$$y_G = \frac{m(L/2) + m(0) + m(L/2)}{3m} = \frac{mL}{3m} = \frac{L}{3}$$

Result: $G\left(\frac{L}{2}, \frac{L}{3}\right)$ *Rationale:* The CoM is horizontally centered, but vertically closer to the bottom because the bottom rod adds mass there.

3 Supplementary Problem 1: Center of Mass of a U-Shaped Plate

Method: Decomposition. We assume the plate is homogeneous with uniform mass density σ . Mass is proportional to area ($m_i \propto A_i$). We decompose the U-shape into three rectangles placed side-by-side along the x -axis.

1. **Left Rectangle ($1\text{m} \times 3\text{m}$):** Area $A_1 = 3 \text{ m}^2$. Center $G_1(0.5, 1.5)$.
2. **Middle Rectangle ($4\text{m} \times 1\text{m}$):** Area $A_2 = 4 \text{ m}^2$. Starts at $x = 1$. Center $G_2(1 + 2, 0.5) = G_2(3, 0.5)$.
3. **Right Rectangle ($1\text{m} \times 2\text{m}$):** Area $A_3 = 2 \text{ m}^2$. Starts at $x = 5$. Center $G_3(5 + 0.5, 1) = G_3(5.5, 1)$.

Total Area $A_{total} = 3 + 4 + 2 = 9 \text{ m}^2$.

Calculations**Abscissa (X_G):**

$$X_G = \frac{A_1x_1 + A_2x_2 + A_3x_3}{A_{total}} = \frac{3(0.5) + 4(3) + 2(5.5)}{9}$$

$$X_G = \frac{1.5 + 12 + 11}{9} = \frac{24.5}{9} \approx 2.72 \text{ m}$$

Ordinate (Y_G):

$$Y_G = \frac{A_1y_1 + A_2y_2 + A_3y_3}{A_{total}} = \frac{3(1.5) + 4(0.5) + 2(1)}{9}$$

$$Y_G = \frac{4.5 + 2 + 2}{9} = \frac{8.5}{9} \approx 0.94 \text{ m}$$

Final Result: $G(2.72 \text{ m}, 0.94 \text{ m})$ **4 Problem 7: External Forces Analysis***Definition:* An external force is exerted by an agent outside the defined system boundary.**a) System: Block A**

- Weight $\vec{W}_A$ (Agent: Earth)
- Tension $\vec{T}_{string}$ (Agent: String)
- Elastic Force $\vec{F}_{spring \rightarrow A}$ (Agent: Spring)

b) System: Block B

- Weight $\vec{W}_B$ (Agent: Earth)
- Elastic Force $\vec{F}_{spring \rightarrow B}$ (Agent: Spring)

Note: The string does not touch B, so it exerts no contact force on it.**c) System: Block A + Spring**

- Total Weight $\vec{W}_{A+S}$ (Earth)
- Tension $\vec{T}_{string}$ (String)
- Elastic Force $\vec{F}_{B \rightarrow spring}$ (Block B)

d) System: Block B + Spring

- Total Weight $\vec{W}_{B+S}$ (Earth)
- Elastic Force $\vec{F}_{A \rightarrow spring}$ (Block A)

e) System: A + B + Spring (Entire hanging assembly)

- Total Weight $\vec{W}_{sys}$ (Earth)
- Tension $\vec{T}_{string}$ (String)

Note: All spring forces and interactions between blocks are now internal.

5 Example 9.4: Acceleration of Center of Mass

System: Two particles m_1 and m_2 . **Forces:** $\vec{f}_1$ (on m_1 , leftwards), $\vec{f}_2$ (on m_2 , rightwards).

Theorem of Center of Mass: The Center of Mass moves as if all mass is concentrated there and all external forces are applied to it.

$$\sum \vec{F}_{ext} = (m_1 + m_2)\vec{a}_G$$

Projection onto the horizontal axis (Ox , positive to right):

$$-f_1 + f_2 = (m_1 + m_2)a_G$$

Final Expression:

$$a_G = \frac{f_2 - f_1}{m_1 + m_2}$$

Interpretation: If $f_2 > f_1$, acceleration is positive (right). If $f_1 > f_2$, acceleration is negative (left).

6 Problem 17: Dynamics of a Train

System: Locomotive + Middle Car + Rear Car. Mass of each = M . Let T_1 be tension between Locomotive and Middle Car. Let T_2 be tension between Middle Car and Rear Car.

a) Case without Friction

1. **Acceleration** (a) Apply Newton's 2nd Law to the whole system ($3M$):

$$F = 3M \cdot a \implies a = \frac{F}{3M}$$

2. **Tension** T_2 Isolate Rear Car (M). Only horizontal force is T_2 :

$$T_2 = M \cdot a = M \left(\frac{F}{3M} \right) \implies T_2 = \frac{F}{3}$$

3. **Tension** T_1 Isolate Middle Car (M). Forces: T_1 (forward), T_2 (back):

$$T_1 - T_2 = M \cdot a \implies T_1 = \frac{F}{3} + \frac{F}{3} \implies T_1 = \frac{2F}{3}$$

b) Case with Friction (f)

Friction f exists on each of the two cars. Total resistive force = $2f$.

1. **New Acceleration** (a') Whole System Equation:

$$F - 2f = 3M \cdot a' \implies a' = \frac{F - 2f}{3M}$$

2. **New Tension** T'_2 Isolate Rear Car (T'_2 forward, f backward):

$$T'_2 - f = Ma' \implies T'_2 = f + M \left(\frac{F - 2f}{3M} \right)$$

$$T_2' = \frac{3f + F - 2f}{3} \Rightarrow \boxed{T_2' = \frac{F + f}{3}}$$

3. New Tension T_1' Isolate Middle Car (T_1' forward, T_2' back, f back):

$$T_1' - T_2' - f = Ma'$$

$$T_1' = T_2' + f + Ma' = \frac{F + f}{3} + f + \frac{F - 2f}{3}$$

$$T_1' = \frac{F + f + 3f + F - 2f}{3} \Rightarrow \boxed{T_1' = \frac{2(F + f)}{3}}$$

7 Problem 18: Dynamics of Two Particles

Givens:

- $m_1 = 1$ kg, starts at $x = 0$. Forces: $\vec{F}_1 = -2\vec{i}$, $\vec{F}_2 = \vec{i}$. Net $\vec{F}_M = -\vec{i}$.
- $m_2 = 2$ kg, starts at $x = 3$. Forces: $\vec{F}_3 = 6\vec{i}$, $\vec{F}_4 = -4\vec{i}$. Net $\vec{F}_N = 2\vec{i}$.

a) Position of Center of Mass at $t = 0$

$$x_G(0) = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2} = \frac{1(0) + 2(3)}{1 + 2} = \frac{6}{3} = \boxed{2 \text{ m}}$$

b) Equations of Motion

Particle M: Acceleration $a_1 = \frac{F_M}{m_1} = \frac{-1}{1} = -1 \text{ m/s}^2$. Using $x(t) = \frac{1}{2}at^2 + v_0t + x_0$ (starts from rest):

$$\boxed{x_M(t) = -0.5t^2}$$

Particle N: Acceleration $a_2 = \frac{F_N}{m_2} = \frac{2}{2} = 1 \text{ m/s}^2$.

$$\boxed{x_N(t) = 0.5t^2 + 3}$$

c) Position of Center of Mass at time t

Substitute $x_M(t)$ and $x_N(t)$ into the CoM formula:

$$x_G(t) = \frac{1(-0.5t^2) + 2(0.5t^2 + 3)}{3}$$

$$x_G(t) = \frac{-0.5t^2 + 1.0t^2 + 6}{3} = \frac{0.5t^2 + 6}{3}$$

$$\boxed{x_G(t) = \frac{1}{6}t^2 + 2}$$

d) **Verification:** $\sum \vec{F}_{ext} = M\vec{a}_G$

LHS (Sum of Forces):

$$\sum \vec{F}_{ext} = \vec{F}_M + \vec{F}_N = (-1\vec{i}) + (2\vec{i}) = 1\vec{i} \text{ N}$$

RHS (Mass $\times$ Accel of CoM): Derive a_G from $x_G(t)$:

$$v_G(t) = \frac{dx_G}{dt} = \frac{1}{3}t$$

$$a_G(t) = \frac{dv_G}{dt} = \frac{1}{3} \text{ m/s}^2 \implies \vec{a}_G = \frac{1}{3}\vec{i}$$

$$M\vec{a}_G = (3 \text{ kg}) \cdot \left(\frac{1}{3}\vec{i}\right) = 1\vec{i} \text{ N}$$

Conclusion: LHS = RHS ($1\vec{i} = 1\vec{i}$). The relation is verified.

8 Supplementary Problem 2: System of 5 Particles

System Data: A system of 5 particles in a 2D coordinate system (x, y in meters).

- $m_1 = 3 \text{ kg}$ at $A(2, 2)$
- $m_2 = 1 \text{ kg}$ at $B(-2, 1)$
- $m_3 = 2.5 \text{ kg}$ at $C(-1, -3)$
- $m_4 = 1.5 \text{ kg}$ at $D(3, -1)$
- $m_5 = 2 \text{ kg}$ at $E(0, 4)$

1. Total Mass M

$$M = \sum m_i = 3 + 1 + 2.5 + 1.5 + 2 = \boxed{10 \text{ kg}}$$

2. Abscissa x_G of Center of Mass

$$x_G = \frac{\sum m_i x_i}{M} = \frac{3(2) + 1(-2) + 2.5(-1) + 1.5(3) + 2(0)}{10}$$

$$x_G = \frac{6 - 2 - 2.5 + 4.5 + 0}{10} = \frac{6}{10} = \boxed{0.6 \text{ m}}$$

3. Ordinate y_G of Center of Mass

$$y_G = \frac{\sum m_i y_i}{M} = \frac{3(2) + 1(1) + 2.5(-3) + 1.5(-1) + 2(4)}{10}$$

$$y_G = \frac{6 + 1 - 7.5 - 1.5 + 8}{10} = \frac{6}{10} = \boxed{0.6 \text{ m}}$$

4. Position of G

The center of mass G is located at coordinates **$G(0.6, 0.6)$** .

Geometric Interpretation: On the diagram, G lies in the first quadrant, relatively close to the origin $O(0, 0)$, slightly biased towards the heavy mass at $A(2, 2)$ and the upper mass $E(0, 4)$, which balance the spread of the other masses. It is marked at $x = 0.6$ and $y = 0.6$.